

# PRAKHAR TRIPATHI

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## EDUCATION

### Stevens Institute of Technology

Master of Science in Computer Science

Hoboken, NJ

Jan 2025 - Dec 2026

- Relevant Coursework: Cloud Systems, Applied Machine Learning, Natural Language Processing, .

### Manipal Institute of Technology

Bachelor of Technology in Computer and Communications Engineering

Manipal, India

Oct 2020 - Jun 2024

- Relevant Coursework: Computer Architecture, Distributed Computing, Object-Oriented Design, Data Structures.

## TECHNICAL PROFICIENCIES

**Languages:** Java, Python, C++, SQL, JavaScript, TypeScript, PHP, MySQL, HTML/CSS.

**Data & AI:** PyTorch, TensorFlow, JAX, Streaming Data Architectures, Machine Learning, Distributed Systems.

**Tools & Automation:** Google Cloud Platform, Deployment Automation, Benchmarking, Git, Linux/Windows, VMware.

## EXPERIENCE

### Software Engineering Intern

Genpact

Hyderabad, India

Feb 2024 - Aug 2024

- Developed automated data processing workflows in Python to interface with large scale enterprise backend infrastructure, optimizing execution pathing for hybrid server deployments.
- Designed and executed performance benchmarking and quality validation testing protocols across multi-node testing environments to evaluate pipeline stability under extreme data load conditions.
- Collaborated tightly with cross-functional development groups to translate statistical data requirements into maintainable, low-latency code integrations within production software structures.

### Software Development Engineer

Accelth Technologies Pvt Ltd

Manipal, India

Jun 2023 - Aug 2023

- Engineered web-based infrastructure tracking modules and operational data management utilities using JavaScript, Python, and SQL to support multi-site environment synchronization.
- Authored custom object-oriented libraries in Java to bridge low-level physical telemetry systems with centralized JVM-based stream tracking interfaces, improving execution monitoring.
- Investigated and debugged end-to-end connectivity bottlenecks across hybrid network layers, driving runtime process simplifications during scheduled system validation ceremonies.

## PROJECTS

### GCP Deployment Automation & Model serving Stack | GCP, Python, Java, PyTorch

- Architected an automated model deployment pipeline on Google Cloud Platform to manage end-to-end model handoff, staging, and live inference serving using hybrid compute stacks.
- Integrated optimized model inference layers with Java-based streaming data pipelines, implementing structural monitoring interfaces to track production runtime metrics and evaluate live model responses.
- Built systematic model sampling routines to feed target subsets back into researcher feedback loops, ensuring continuous quality evaluation against baseline parameters.

### Fake News Detection & Content Analysis System | Python, DeBERTa, PyTorch, Transformers

- Engineered an end-to-end natural language processing pipeline using DeBERTa to automatically analyze, identify, and classify misinformation and fake news across large-scale text repositories.
- Fine-tuned deep transformer architectures using PyTorch and Hugging Face libraries, implementing custom text sanitization, dynamic batching, and optimized tokenization mechanisms to process unstructured data.
- Optimized inference workloads to reduce latency, achieving a 96.4% F1-score on text classification benchmark.

### StoryLine Adaptive Real-Time Processing System | Distributed Systems, Swift, Data Architectures

- Developed an agile real-time pipeline to handle high-throughput, unstructured data processing requests, rapidly adapting code to match evolving hybrid infrastructure requirements.
- Implemented distributed debugging protocols to validate system integrations end-to-end, guaranteeing flawless execution flows from ingestion layers down to data serving endpoints.

## AWARDS & ACCOMPLISHMENTS

Won InnovateFest 22 at MIT Manipal by showcasing our company Accelth product.

Secured an extraordinary 442nd World Rank globally in the Forensics category at ArmstrongCTF'22.

Recognized as a Top 2% Global Intern among 700+ participants in Genpact's selective NextGen builder program.